Lecture 4: Neural Networks

Introduction to Machine Learning, week 4 notes

1. From linear models to neurons

A single artificial neuron computes a weighted sum of its inputs, adds a bias, and passes the result through an activation function: $y = f(w \cdot x + b)$. With the identity function this is just linear regression; with the logistic sigmoid it becomes logistic regression.

Stacking neurons into layers lets the network learn non-linear decision boundaries. A network with one hidden layer of enough units can approximate any continuous function on a bounded region (the universal approximation theorem), although it may need a very large number of units.

2. Activation functions

- **Sigmoid**: squashes values to (0, 1). Saturates for large inputs, which slows learning (vanishing gradients).
- **Tanh**: output in (−1, 1) and zero-centred, but it also saturates.
- **ReLU**: max(0, z). Cheap and does not saturate for positive inputs; the default choice for hidden layers.
- **Softmax**: turns the output layer into probabilities that sum to 1 for multi-class problems.

3. Training with backpropagation

Training minimises a loss such as cross-entropy. Backpropagation applies the chain rule layer by layer to compute the gradient of the loss with respect to every weight, and gradient descent updates each weight by $w \leftarrow w - \eta \, \partial L / \partial w$, where η is the learning rate.

4. Practical tips

- Initialise weights randomly (He initialisation for ReLU) so that neurons do not all learn the same thing.
- Use mini-batches of 32 to 256 examples; a learning rate of 0.001 with the Adam optimiser is a good starting point.
- Watch the validation loss: if it rises while training loss falls, the model is overfitting. Add dropout (p = 0.2 to 0.5), weight decay, or more data.
- Normalise inputs to zero mean and unit variance; batch normalisation helps deeper networks train faster.

5. Worked example

On the MNIST handwritten digits dataset (60,000 training images of 28 × 28 pixels), a network with two hidden layers of 128 ReLU units reaches about 98% test accuracy after 10 epochs, compared with about 92% for plain logistic regression.

6. For next week

- Read chapter 6 of the course textbook (Deep feedforward networks).
- Problem set 3 is due Friday 13 November at 23:59.
- Install PyTorch before the lab session on Tuesday.